(0 Points) Examine the following C code snippet carefully:

```
void process_data(const char* input) {
char buffer[16];
strncpy(buffer, input, sizeof(buffer));
printf("DataProcessed:%s\n", buffer);
}
```

If the function is called with a string `input` of length 20 (e.g., `"ThisIsASecretMessage"`), which of the following statements is **TRUE**?

A. The code is completely safe because `strncpy` prevents writing beyond the 16th byte of `buffer`.

B. A Stack Buffer Overflow occurs because `strncpy` will force a null-terminator at `buffer[16]`.

C. A Buffer Over-read may occur during `printf` because `buffer` will not be null-terminated.

D. The code will trigger a segmentation fault during `strncpy` because the input is stored in read-only memory.

(0 Points) Alice and Bob agree to use the textbook Diffie-Hellman key exchange protocol with the public parameters $p = 31$ (modulus) and $g = 3$ (generator).

(a) Alice chooses her private key $a = 12$. Calculate her public key $A = g^a \pmod{p}$.

(b) Bob chooses his private key $b = 7$. Calculate his public key $B = g^b \pmod{p}$.

(c) Alice receives B and computes the shared secret $K = B^a \pmod{p}$. Calculate K.

(0 Points) Consider the multiplicative group $G = (\mathbb{Z}_{55}^*, \cdot)$.

(a) True or False: The group G is a cyclic group.

(b) What is the order of the group $|G|$?

(c) True or False: The element $g = 2$ is a generator of G.

(d) Calculate $2^{126} \pmod{55}$. Please express all answers in this assignment involving modular arithmetic (e.g., $\pmod{n}$) as the smallest integer in the range $[0, n-1]$.

(0 Points) **True or False:** Let $p = 23$. $g = 5$ is a generator of $\mathbb{Z}_{23}^*$. Alice and Bob perform a Diffie-Hellman exchange. An eavesdropper intercepts $A = g^a = 8$ and $B = g^b = 10$. They are also given a challenge value Z, which is either $Z_0 = g^{ab} \pmod{23}$ or $Z_1 \xleftarrow{\$} \mathbb{Z}_{23}^*$. If the eavesdropper receives $Z = 7$, they can determine with 100% certainty that Z is a random element (Z_1) by checking the Legendre symbol $(\frac{x}{p}) \equiv x^{(p-1)/2} \pmod{p}$.

(0 Points) **True or False:** Consider a standard RSA-CRT implementation where the decryption (or signing) process is optimized using the Chinese Remainder Theorem to compute $M^d \pmod{n}$. The RSA-CRT construction is IND-CPA secure.

(0 Points) Let $p = 197$ and consider the multiplicative group $G = (\mathbb{Z}_{197}^*, \cdot)$. We want to find the smallest positive integer x such that:

$$2^x \equiv 14 \pmod{197}$$

(a) True or False: The order of the element $g = 2$ in $\mathbb{Z}_{197}^*$ is 196.

(b) Calculate $2^{98} \pmod{197}$.

(c) Calculate $2^{50} \pmod{197}$.

(d) What is the value of x?

(0 Points) Let $G : \{0,1\}^n \to \{0,1\}^{\ell(n)}$ be a secure PRG where $\ell(n) > 2n$. Bob attempts to construct a new PRG $G_{new} : \{0,1\}^n \to \{0,1\}^{2\ell(n)}$. The output is defined as:

$$G_{new}(s) := G(s) \parallel (G(s \oplus 1^n) \oplus G(s)).$$

Is G_{new} a secure PRG?

- **A. Yes.** The security follows from the security of G via a hybrid argument. If an adversary could distinguish $G_{new}(s)$ from a random string $r_1 \parallel r_2$, they would necessarily be able to distinguish $G(s)$ from a random string r_1, which contradicts the assumption that G is secure.
- **B. Yes.** Since s and $s \oplus 1^n$ are distinct seeds, the outputs $G(s)$ and $G(s \oplus 1^n)$ are computationally independent and pseudorandom. XORing them with each other further "shuffles" the entropy, maintaining security.
- **C. No.** An adversary $\mathcal{D}$ can distinguish the output by receiving $y_1 \parallel y_2$, then computing $y_1 \oplus y_2$.
- **D. No.** The output length $2\ell(n)$ is larger than the stretch allowed for a secure PRG derived from a seed of length n, violating the expansion factor limits of the standard model.

(0 Points) A developer is implementing a login feature and writes the following pseudocode to prevent SQL injection by manually escaping single quotes:

```
$username = $_POST['username'];
$password = $_POST['password'];

$safe_username = str_replace("'", "''", $username);

$sql = "SELECT * FROM users
        WHERE username = '$safe_username'
        AND password = $password";

$result = $db->execute($sql);
```

Which of the following statements best describes the security of this code?

A. The code is secure because all single quotes in the username are escaped, preventing an attacker from breaking out of the string literal.

B. The code is vulnerable to SQL injection because the `$password` variable is not enclosed in single quotes and is not sanitized.

C. The code is vulnerable to "Second-Order SQL Injection" because the escaped string is stored in the database.

D. The code is secure against SQL injection but vulnerable to Cross-Site Scripting (XSS) due to the use of `str_replace`.

(0 Points) Alice uses the ElGamal encryption scheme with the public parameters $p = 197$ (modulus) and $g = 2$ (generator). Her public key is $h = 14 \pmod{197}$.

(a) Calculate Alice's private key x, which is the smallest positive integer satisfying $g^x \equiv h \pmod{197}$.

(b) Bob wants to send a message $m = 50$ to Alice. He chooses a random key $k = 98$. Calculate the ciphertext (c_1, c_2), where:

$$c_1 = g^k \pmod{197}, \quad c_2 = m \cdot h^k \pmod{197}$$

(c) The decryption process involves calculating the shared secret $s = c_1^x \pmod{197}$. Calculate s.

(0 Points) An inexperienced engineer decides to use $e = 2$ as the public exponent for a textbook RSA system to speed up encryption, with the public modulus $n = 437$.

(a) Explore why this system is not a valid RSA cryptosystem by calculating $\gcd(e, \phi(n))$. (Note: $n = 19 \times 23$).

(b) Given a ciphertext $C = 139$, the attacker wants to recover the plaintext M. Calculate the smallest positive one among the possible plaintexts. (Hint: For $x^2 \equiv a \pmod{p}$, if $p \equiv 3 \pmod{4}$, then $x \equiv \pm a^{(p+1)/4} \pmod{p}$).

(c) From the second question, we know incorrectly set $e = 2$ will break the 1-to-1 mapping between the plaintext and the ciphertext. Suppose the engineer restarts the scheme with $e = 2, n' = 323$. An attacker obtains two possible plaintexts $M_1 = 4$ and $M_2 = 72$ when decrypting a ciphertext. Help the attacker to recover p, q such that $p * q = n' = 323$ and $p < q$.